APPSC INTEGRATED SUBJECT INDEX & MICRO-TOPIC OVERLAP ANALYZER

# 04, 12, 37 & 41B. SCIENCE, TECHNOLOGY, ICT, ENERGY & BIOTECH

Underlined Highlights: **Group-2 Mains Extras** | **Group-1 Mains P-V Extras** | **Group-1 Prelims Sec-B Extras** | **Common GS / Draughtsman Extras**

GR-2 MAINS: P-II SEC-B (Units 1–5 | 75M) GR-1 MAINS: PAPER-V (Units 1–9 | 150M)

GR-1 PRELIMS: P-II SEC-B (Units 18–22 | 40M) COMMON GS P-I (Item 2) & DRAUGHTSMAN P-I (U1)

| Unit | Consolidated Theme Title | Exams & Notification Clause Mapping | Overlap Type | Page |
|---|---|---|---|---|
| 01 | **S&T Policies, Missions, Institutes & IPR** | GR2 U1 • GR1 Prelims U18 • GR1 Mains P-V U1, U9 | Core STI Policy + GR1 Institutes/IPR | 01 |
| 02 | **Space Technology (ISRO) & Defence (DRDO)** | GR2 U1 • GR1 Prelims U20 • GR1 Mains P-V U3 | IGMDP (GR2) vs IRNSS/Edusat (GR1) | 02 |
| 03 | **ICT, Digital India, E-Governance & Cyber Security** | GR2 U1 • GR1 Prelims U19 • GR1 Mains P-V U2 | Netiquettes (GR1P) vs AP IT Dev (GR1M) | 03 |
| 04 | **Nuclear Technology & Energy Management** | GR2 U1, U2 • GR1 Prelims U21 • GR1 Mains P-V U4 | BS Norms/Biofuels (GR2) vs Int'l Nuclear (GR1) | 04 |
| 05 | **Biotechnology, Nanotech & Agri-Revolutions** | GR2 U4 • GR1 Prelims U22 • GR1 Mains P-V U6, U7, U8 | Transgenics/Biorem (GR2) vs Nanotech/Tissue (GR1) | 05 |
| 06 | **Human Health, Diseases, Vaccines & Everyday Science** | GR2 U5 • GR1 Mains P-V U8 • Common GS Item 2 • DM U1 | Healthcare Delivery (GR2) vs Pathogens/Vaccines (GR1) | 06 |

## UNIT 01 • NATIONAL S&T POLICIES, MISSIONS, SCIENTIFIC INSTITUTES & IPR

GR2 MAINS U1 | GR1 PRELIMS U18 | GR1 MAINS P-V U1, U9

**1. Technology Missions, Policies and Applications (Policy Portion)** GR-2 MAINS P-II (75M)

National S&T Policy: Recent Science, Technology and Innovation Policy, and **National Strategies and Missions, Emerging Technology Frontiers**.

**18. Science and Technology (Group-1 Screening Paper-II Sec-B)** GR-1 PRELIMS P-II (SEC-B)

Nature and Scope of Science & Technology; Relevance of Science & Technology to the day-to-day life; National Policy on Science, Technology and Innovation; **Institutes and Organization in India promoting integration of Science, Technology and Innovation, their activities and contribution; Contribution of Prominent Indian Scientists**.

**1 & 9. Integration of S&T, Institutes, Scientists & IPR** GR-1 MAINS PAPER-V

**Unit 1:** Integration of Science, Technology and Innovation for better human life; S&T in everyday life; National Policies on proliferation of S&T and Innovation; India's contribution in the field of S&T. **Concerns and challenges in the proliferation and use of S&T; Role and Scope of S&T in nation building. Major Scientific institutes for S&T in AP and India. Major Scientific Institutes for R&D in AP and India. Achievements of Indian Scientists – Indigenous technologies and developing new technologies**.

**Unit 9:** **Issues related to Intellectual Property Rights (IPR) in the field of Science and Technology. Promotion of Science in AP and India**.

## UNIT 02 • SPACE TECHNOLOGY (ISRO) & DEFENCE TECHNOLOGY (DRDO)

GR2 MAINS U1 | GR1 PRELIMS U20 | GR1 MAINS P-V U3

**1. Space Technology & Defence Technology** — GR-2 MAINS P-II (75M)

**Space Technology:** Launch Vehicles of India, Recent Indian Satellite Launches and its applications, Indian Space Science Missions.
**Defence Technology:** Defence Research and Development Organization (DRDO): Structure, Vision and Mission, Technologies Developed by the DRDO, Integrated Guided Missile Development Programme (IGMDP).

**20. Technology in Space & Defence** — GR-1 PRELIMS P-II (SEC-B)

Evolution of Indian Space Programme; Indian Space Research Organization (ISRO) – its activities and achievements; Various Satellite Programmes – Satellites for Telecommunication, Indian Regional Navigation Satellite System (IRNSS / NavIC), Indian Remote Sensing (IRS) Satellites; Satellites for defence, Edusat or Satellites for academic purposes; Defence Research and Development Organization (DRDO) – vision, mission and activities.

**3. Indian Space Programme & DRDO** — GR-1 MAINS PAPER-V

Indian Space Programme – Past, Present and Future; Indian Space Research Organization (ISRO) – its activities and achievements; Satellite Programmes of India and Use of Satellites in different fields like Health (Telemedicine), Education (Tele-education), Communication Technology, Weather forecasting affecting human lives; Defence Research and Development Organization (DRDO).

| Unit 02: 100% Common Core (Gr-2 U1 + Gr-1 Prelims & Mains) | Explicitly Detailed in Group-2 Mains (Unit 1) | Explicitly Detailed in Group-1 (Prelims U20 & Mains U3) |
|---|---|---|
| • ISRO Organization, Space Commission, Antrix, NSIL & IN-SPACe (Indian Space Policy 2023)<br>• Communication (INSAT/GSAT/CMS) & Earth Observation / Remote Sensing (IRS/EOS) Satellites<br>• DRDO Organization, Vision, Mission & Labs (including NSTL Visakhapatnam) | • Generations of ISRO Launch Vehicles: SLV-3, ASLV, PSLV, GSLV Mk-II, LVM3 (Cryogenic CE-20), SSLV & NGLV (Soorya)<br>• Indian Space Science & Interplanetary Missions (Chandrayaan 1/2/3/4, Mangalyaan, AstroSat, Aditya-L1, XPoSat, Gaganyaan, Shukrayaan, Bharatiya Antariksh Station)<br>• IGMDP (Prithvi, Agni, Trishul, Nag, Akash) & DRDO Weapon Systems (BrahMos, Pralay, Astra, Pinaka, Arjun, Tejas, Ballistic Missile Defence, Mission Divyastra MIRV) | • IRNSS / NavIC (7-satellite regional navigation constellation & NVS series)<br>• Dedicated Defence Satellites (GSAT-7 Rukmini, GSAT-7A Angry Bird, EMISAT) & Edusat (GSAT-3)<br>• Socio-Economic Applications of Satellites in Tele-medicine, Tele-education, Disaster Warning & Meteorological Forecasting |

⚡ **YUDDAM 2027 AUDIT INSIGHT (UNIT 02):** Notice how Group-2 Unit 1 explicitly names Launch Vehicles, Space Science Missions & IGMDP while Group-1 Prelims U20 explicitly names IRNSS, IRS, Defence Satellites & Edusat. In actual APPSC papers, **both sets are 100% interchangeable**—read them as a single consolidated Space + Defence block!

## UNIT 03 • INFORMATION & COMMUNICATION TECHNOLOGY (ICT) & CYBER SECURITY

GR2 MAINS U1 | GR1 PRELIMS U19 | GR1 MAINS P-V U2 (AND P-III C15)

**1. Information and Communication Technology (ICT)** — GR-2 MAINS P-II (75M)

Information and Communication Technology (ICT): National Policy on Information Technology – **Digital India Mission: Initiatives and Impact** – E-Governance Programmes and Services – Cyber Security concerns – National Cyber Security Policy.

**19. Information and Communication Technology (ICT)** — GR-1 PRELIMS P-II (SEC-B)

Nature and Scope of ICT; ICT in day-to-day life; **ICT and Industry**; ICT and Governance – Various government schemes promoting use of ICT, E-Governance programmes and services; **Netiquettes**; Cyber Security Concerns – National Cyber Crime Policy.

**2. Information and Communication Technology (ICT)** — GR-1 MAINS PAPER-V

Information and Communication Technology (ICT) – its importance, advantages and **challenges**; E-governance and India; **Cyber Crime** and policies to address security concerns. Government of India Policy on Information Technology (IT). **IT Development in AP and India**.

| Unit 03: 100% Common Core (Gr-2 U1 + Gr-1 Prelims & Mains + Common GS) | Extra in Group-2 Mains & Group-1 Prelims | Extra in Group-1 Mains (P-V U2 & P-III C15) |
|---|---|---|
| • Basics of ICT, Spectrum bands, Optical Fiber, Broadband, Cloud Computing & IndiaAI<br>• National Policy on IT (2012) & National Policy on Software Products (2019)<br>• National e-Governance Plan (NeGP), e-Kranti (NeGP 2.0), UMANG, DigiLocker, PRAGATI, BharatNet, India Stack / DPI<br>• Cyber Threats (Malware, Ransomware, Phishing, Zero-Day, DDoS, Deepfakes), CERT-In, NCIIPC & National Cyber Security Policy (2013 / updated strategy) | • **Digital India Mission (9 Pillars, Bhashini, ONDC, DIKSHA, e-Sanjeevani, PM-WANI)**<br>• **Netiquettes (Internet etiquette, digital citizenship, social media guidelines)**<br>• **ICT in Industry (Industry 4.0, Smart Manufacturing, Automation)** | • **IT Development in Andhra Pradesh (Visakhapatnam IT Hub, Tirupati EMC, AP IT & Global Capability Centres Policy, AP FiberNet, Mana Mitra WhatsApp Governance)** (Note: This also overlaps with Group-2 Economy Unit 5!)<br>• **IT Act 2000 (Amendments), DPDP Act 2023 & Jurisdictional challenges in Cyber Crimes (Cross-referenced with Gr-1 Mains Paper-III C15)** |

## UNIT 04 • NUCLEAR TECHNOLOGY & CLEAN ENERGY MANAGEMENT

GR2 MAINS U1, U2 | GR1 PRELIMS U21 | GR1 MAINS P-V U4

**1 (Nuclear Portion) & 2 (Energy Management)** — GR-2 MAINS P-II (75M)

**Unit 1 (Nuclear):** Nuclear Technology: Indian Nuclear Reactors and Nuclear Power Plants – Applications of Radioisotopes – India's Nuclear Programme.
**Unit 2 (Energy Management):** Policy and Projections: Installed Energy Capacities and Demand in India – National Energy Policy – National Policy on Biofuels – Bharat Stage Norms – Non-Renewable and Renewable Energy: Sources and Installed Capacities in India – New Initiatives and Recent Programmes, Schemes and Achievements in India's Renewable Energy Sector.

**21. Energy Requirement and Efficiency** — GR-1 PRELIMS P-II (SEC-B)

India's existing energy needs and deficit; India's Energy Resources and Dependence, Energy policy of India – Government Policies and Programmes. Solar, Wind and Nuclear energy.

**4. India's Energy Needs, Efficiency, Clean Energy & Nuclear Policy** — GR-1 MAINS PAPER-V

India's energy needs, efficiency and resources; Clean energy resources; Energy policy of India – Government Policies and Programmes. Conventional and Non-Conventional energy resources. Energy demands, Indian Energy Sciences, Conventional energy powers, Thermal, renewable energy resources: Solar, Wind, Bio and waste-based, energy policies, Geothermal and Tidal Sources, energy Policies in India, energy security. Salient features of Nuclear Policy of India; Development of Nuclear programmes in India, Nuclear Policies at the International level and India's stand on them.

| Unit 04: 100% Common Core (Gr-2 U1,U2 + Gr-1 Prelims & Mains) | Extra Sub-Topics ONLY in Group-2 Mains (U1, U2) | Extra Sub-Topics ONLY in Group-1 (Prelims U21 & Mains U4) |
|---|---|---|
| • Homi Bhabha's 3-Stage Indian Nuclear Power Programme (PHWRs, Fast Breeder Reactors / PFBR Kalpakkam, Thorium/U-233 AHWR)<br>• Operating & Planned Nuclear Power Plants in India (including Kovvada in AP)<br>• Conventional (Coal/Thermal, Hydro, Gas) vs Renewable (Solar, Wind, Biomass, Small Hydro) energy sources & National Energy Policy | • Applications of Radioisotopes in Agriculture (BARC mutant varieties, food irradiation), Medicine (Co-60, I-131, Tc-99m), Industry & Hydrology<br>• National Policy on Biofuels (2018 & 2022 Amendments: E20 Ethanol blending, 1G–4G biofuels, Global Biofuels Alliance)<br>• Bharat Stage (BS-VI) Emission Norms & Fuel Standards<br>• Exact Installed Capacity Figures (GW/MW share of Coal, Solar, Wind, Hydro, Nuclear) & MNRE Schemes (PM Surya Ghar, PM-KUSUM, National Green Hydrogen Mission) | • India's Nuclear Doctrine (1999/2003: No First Use, Credible Minimum Deterrence, Nuclear Command Authority)<br>• International Nuclear Treaties & Export Regimes: NPT, CTBT, FMCT, NSG, MTCR, Wassenaar, Australia Group, IAEA Safeguards & 123 Civil Nuclear Deals<br>• Waste-to-Energy, Geothermal (Puga Valley, Manikaran) & Tidal/Wave/OTEC Energy<br>• Energy Efficiency (BEE, PAT Scheme, ECBC, Star Labeling) & Energy Security / Import Dependence |

## UNIT 05 • BIOTECHNOLOGY, GENETIC ENGINEERING, NANOTECH & AGRI-TECH

GR2 MAINS U4 | GR1 PRELIMS U22 | GR1 MAINS P-V U6, U7, U8

**4. Waste Management and Pollution Control (Agri-Biotech Portion)** — GR-2 MAINS P-II (75M)

Impact of Transgenics on Environment and their Regulation – Eco-Friendly Technologies in Agriculture – Bioremediation: Types and Scope in India.

**22. Environmental Science (Biotech & Nanotech Portion)** — GR-1 PRELIMS P-II (SEC-B)

Biotechnology and Nanotechnology; Nature, Scope and application, Ethical, Social, and Legal issues, Government Policies. Genetic Engineering; Issues related to it and its impact on human life.

**6 (Revolutions), 7 (Biotech & Nanotech) & 8 (Genetic Engg & Agri-Biotech)** — GR-1 MAINS PAPER-V

**Unit 6 (End):** White Revolution, Green Revolution, and Green Pharmacy.

**Unit 7:** Nature, Scope and Applications of Biotechnology and Nanotechnology in India; Ethical, Social and Legal concerns, Government policies; Genetic engineering, issues related to it and its impact on human life. Biodiversity, fermentation, Immuno-diagnosis techniques.

**Unit 8 (Biotech Portion):** Introduction to Genetic Engineering and Biotechnology. Basic concepts of genetic engineering. Tissue culture methods and applications. Biotechnology in agriculture – Bio-pesticides, Bio-fertilizers, Bio-fuels, Genetically modified crops. Animal husbandry – transgenic animals.

| Unit 05: 100% Common Core (Gr-2 U4 + Gr-1 Prelims & Mains) | Extra in Group-2 Mains (Unit 4) | Extra in Group-1 (Prelims U22 & Mains U6, U7, U8) |
|---|---|---|
| • Recombinant DNA Technology, CRISPR-Cas9 Gene Editing & Genetically Modified (GM) Crops (Bt Cotton, Bt Brinjal, DMH-11 Mustard)<br>• Biosafety Regulation in India (GEAC under EPA 1986, DBT, Cartagena Protocol)<br>• Eco-friendly Agriculture: Bio-fertilizers (Rhizobium, Azotobacter, Azolla, Mycorrhiza, Blue-Green Algae) & Bio-pesticides (Bacillus thuringiensis, Trichoderma, Neem) | • Bioremediation: Types (Bioventing, Biosparging, Bioaugmentation, Phytoremediation, Mycoremediation, Oilzapper) & Scope in India (Note: This also overlaps with AEE PCB Unit 8!) | • Nanotechnology: Nature, Scope, Carbon Nanotubes, Graphene, Quantum Dots, Drug Delivery, Nano-Urea/DAP & National Nano Mission<br>• Plant Tissue Culture (Totipotency, Micropropagation, Somatic Hybridization), Fermentation Technology & Immuno-diagnosis (ELISA, PCR, RT-PCR, Western Blot, Monoclonal Antibodies)<br>• Transgenic Animals, Cloning, Stem Cells, Ethical/Legal (ELSI) issues & Green Pharmacy |

## UNIT 06 • HUMAN HEALTH, INFECTIOUS DISEASES, VACCINES & GENERAL SCIENCE

GR2 MAINS U5 | GR1 MAINS P-V U8 | COMMON GS P-I (ITEM 2) | DRAUGHTSMAN P-I (U1)

**5. Environment and Health (Health Issues Portion)** — GR-2 MAINS P-II (75M)

Health Issues: Recent Trends in Disease Burden and Epidemic and Pandemic Challenges in India – Preparedness and Response: Healthcare Delivery and Healthcare Outcomes in India.

**8. Human Diseases, Microbial Infections & Vaccines** — GR-1 MAINS PAPER-V

Human diseases – microbial infections. Common infections and preventive measures. Introduction to bacterial, viral, protozoal and fungal infections. Basic knowledge of infections – diarrhoea, dysentery, cholera, tuberculosis, malaria, viral infections like HIV, Encephalitis, Chikungunya, bird flu – preventive measures during outbreaks. Vaccines: Introduction to immunity, Fundamental concepts in vaccination, Production of Modern Vaccines (production of Hepatitis vaccine).

**2. Common GS Paper-I (150M) & 1. Draughtsman Paper-I (SSC Standard)** — COMMON GS & DRAUGHTSMAN

**Common GS P-I (AEE Civil/PR/PCB, ASWO, EO):** 2. General Science and its applications to the day to day life; Contemporary developments in Science & Technology and Information Technology.

**Draughtsman P-I Unit 1:** 1. General Science – Contemporary developments in S&T and their implications including matters of everyday observation and experience, as may be expected of an individual who has passed Intermediate.

| Unit 06: Common Health & Disease Core | Extra in Group-2 Mains (U5) vs Group-1 Mains (U8) | Extra ONLY in Common GS P-I & Draughtsman P-I |
|---|---|---|
| • Communicable vs Non-Communicable Diseases (NCDs)<br>• Epidemic & Pandemic Outbreaks (COVID-19, Nipah, Zika, Monkeypox/Mpox, Avian Flu H5N1)<br>• National Health Mission (NHM), Universal Immunization Programme (Mission Indradhanush) & Ayushman Bharat / Aarogyasri | • **Group-2 Mains (U5):** Macro Healthcare Metrics (IMR, MMR, TFR, DALYs, Disease Burden Shift from Communicable to NCDs) & Primary/Secondary/Tertiary Healthcare Delivery in India<br>• **Group-1 Mains (U8):** Specific Pathogens (Diarrhoea, Dysentery, Cholera, TB, Malaria, HIV, Encephalitis, Chikungunya, Bird Flu) & Immunology / Recombinant Hepatitis-B Vaccine Production | • Everyday General Science (Physics, Chemistry & Biology): Light/Optics, Sound, Electricity, Heat, Acids/Bases/Salts, Metals/Alloys, Polymers/Soaps, Human Physiology (Digestive, Circulatory, Nervous, Endocrine systems), Vitamins/Deficiency diseases & Plant Physiology<br>*(Tested in Common GS Paper-I for AEE/ASWO/EO/Draughtsman; NOT in Group-2 Mains!)* |

🎯 **MASTER STUDY SEQUENCE FOR SCIENCE & TECHNOLOGY (8-EXAM EXECUTION PLAN):**
**Step 1 (Read Group-2 Mains Unit 1 & Unit 2 First):** Mastering *STI Policy, ISRO, DRDO, ICT/Digital India, Cyber Security, Nuclear Energy, and Renewable Energy* covers **100% of Group-2 Mains S&T (Units 1–2), 80% of Group-1 Prelims (Sec-B), 70% of Group-1 Mains (Paper-V), and the "Contemporary S&T + IT" half of Common GS Paper-I.**
**Step 2 (Group-1 Prelims & Mains Add-On):** Add **Nanotechnology, Plant Tissue Culture, Immuno-diagnosis (ELISA/PCR), IPR, Specific Infectious Diseases & Recombinant Hepatitis Vaccine production.**
**Step 3 (Common GS Paper-I & Draughtsman Add-On):** Revise an NCERT Class 6–10 **Everyday General Science (Physics, Chemistry, Human Biology & Vitamins)** capsule specifically for AEE, ASWO, EO, and Draughtsman Paper-I.